

Session 13: Beyond Dense Transformers

Sparsity, State Space Models, and Architectural Synthesis

Jonathan Lys

IMT Atlantique
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IMT Atlantique
Bretagne - Pays de la Loire
École Mines-Télécom



Training Language Models: From First Principles to Efficient Serving

Lecture Plan

① Architectural Limits of Dense Attention

Outline

1 Architectural Limits of Dense Attention

The Scaling Frontier: Sparsity & Alternative Attention

- **Dense Transformer constraints:**
Every parameter is activated for every token ($\mathcal{O}(N)$ compute cost), and KV-cache footprint grows linearly with context length T .
- **Sparse Mixture-of-Experts (MoE):**
Route each token to Top- k of E specialized MLP experts. Dramatically expands parameter capacity without increasing active FLOPs.
- **Recurrence & State Space Models:**
Mamba / RWKV replace softmax attention with linear recurrent states, achieving $\mathcal{O}(1)$ inference memory per token.

